**vi. Water**: Adequate amount of water in the soil is a pre-requisite for optimum plant growth. Excessive loss of water through transpiration leads to wilting. In general, there are three types of wilting as follows,

***a. Incipient wilting***: Water content of plant cell decreases but the symptoms are not visible.

***b. Temporary wilting***: On hot summer days, the freshness of herbaceous plants reduces turgor pressure at the day time and regains it at night.

***c. Permanent wilting***: The absorption of water virtually ceases because the plant cell does not get water from any source and the plant cell passes into a state of permanent wilting.

**2. Internal factors**

**i. Leaf area**: If the leaf area is more, transpiration is faster and so xerophytes reduce their leaf size.

**ii. Leaf structure**: Some anatomical features of leaves like sunken stomata, the presence of hairs, cuticle, the presence of hydrophilic substances like gum, mucilage help to reduce the rate of transpiration. In xerophytes the structural modifications are remarkable. To avoid transpiration, as in ***Opuntia*** the stem is flattened to look like leaves called **Phylloclade**. **Cladode** or **cladophyll** in ***Asparagus*** is a modified stem capable of limited growth looking like leaves. In some plants, the petioles are flattened and widened, to become **phyllodes** example ***Acacia melanoxylon.***

### 11.6.5 Plant Antitranspirants

The term **antitranspirant** is used to designate any material applied to plants for the purpose of retarding transpiration. An ideal antitranspirant checks the transpiration process without disturbing the process of gaseous exchange. Plant antitranspirants are two types:

**1. To act as a physical barrier above the stomata**

**Colourless plastics, Silicone oil** and **low viscosity waxes** are sprayed on leaves forming a thin film to act as a physical barrier (for transpiration) for water but permeable to $CO_2$ and $O_2$. The success rate of a physical barrier is limited.

**2. Induction of Stomata closure**

**Carbon-di-oxide** induces stomatal closure and acts as a natural antitranspirant. Further, the advantage of using $CO_2$ as an antitranspirant is its inhibition of photorespiration. **Phenyl Mercuric Acetate** (PMA), when applied as a foliar spray to plants, induces partial stomatal closure for two weeks or more without any toxic effect. Use of **abscisic acid** highly induces the closing of stomata. **Dodecenyl succinic acid** also effects on stomatal closure.

**Uses:**

- Antitranspirants reduce the enormous loss of water by transpiration in crop plants.
- Useful for seedling transplantations in nurseries.

### 11.6.6 Guttation

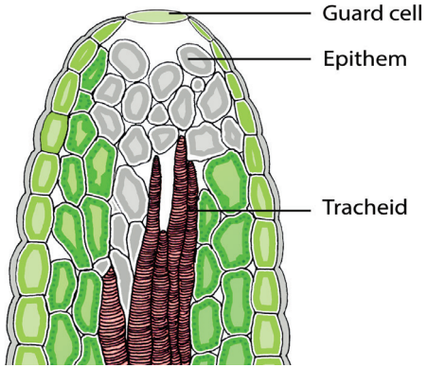



**Figure 11.18:** Structure of Hydathode

During high humidity in the atmosphere, the rate of transpiration is much reduced. When plants absorb water in such a condition root pressure is developed due to excess water within the plant. Thus excess water exudates as liquid from the edges of the leaves and is called **guttation**. Example: Grasses, tomato, potato, brinjal and *Alocasia*. Guttation occurs through stomata like pores called **hydathodes** generally present in plants that grow in moist and shady places. Pores are present over a mass of loosely arranged cells with large intercellular spaces called **epithem**



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